

Mohammed Mowla

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Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C#, HTML/CSS, PHP

Frameworks: Selenium, PyTest, Jakarta EE, JUnit, React, Node.js, Angular, Svelte, Tailwind CSS, JavaFX

Developer Tools: Git/GitHub, AWS S3, Apache Tomcat, MySQL

Libraries: Three.js, p5.js, NumPy, Tailwind CSS, Matplotlib, SciPy, PyTorch

Education

Stockton University

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.78 | Honors: Stockton University Provost's Scholarship (4 Years)

May 2024

Galloway, NJ

Experience

Computer Scientist - Software Automation

Oct. 2024 – Present

LS Technologies: Computer Scientist – A.M.M.S.

Egg Harbor Township, NJ

- Developed and maintained a **Python**, **Selenium**, and **Appium** regression test suite for the **TAP Connected** and **Mobile** applications, safeguarding core processes through ongoing **RMLS** development
- Engineered a **CI/CD** pipeline that automatically pulls, installs, and points the test framework to each new development build via a single command, eliminating manual setup for up to **15** releases per day
- Built automation tooling that generates **Jira** information matrices to quickly analyze team assignments and status to aid in syncing team efforts and focus
- Employed knowledge of **AMMS/RMLS** functionality, including systems: **EM**, **Peabody**, **TAP Connected**, and **Mobile**, to design test cases aligned with real-world user scenarios within an **Agile** environment
- Validated functionality with Subject Matter Experts (**SMEs**) and shared findings with development to drive product improvements

Computer Scientist Intern

June 2024 – Aug. 2024

Federal Aviation Administration – A.M.M.S.

Egg Harbor Township, NJ

- Optimized the automated **TAP iOS** regression suite by refining test scripts and simulator configurations, reducing automation run times.
- Led **11** interns in the development of the Air Traffic Safety Management Team's (AJI-3320) organizational **Wiki** using **Confluence** and **Jira** integrations to improve knowledge sharing
- Expanded automated test coverage of real-world user workflows for **TAP Connected** and **Mobile**, refining procedures to align with team Test and Evaluation (**T&E**) practices.

Computer Scientist Intern

June 2023 – Aug. 2023

Federal Aviation Administration – MSI Gateways Program

Egg Harbor Township, NJ

- Led the initial implementation of **Appium** automation for the **TAP Mobile iOS** application in **Java**, establishing a regression test suite to ensure the continued functionality of critical user scenarios
- Co-led a cohort project researching virtualization of test systems, mentoring peers, and substantially accelerating lab configuration
- Deployed and benchmarked the Simulated Driver Radar Recorder (**SDRR**) and its ecosystem on a cloud-hosted **Ubuntu** instance via **MobaXterm**

Research

Lake Fred Research Project

Jan. 2023 – Nov. 2023

- Gathered geospatial data from Lake Fred at Stockton University and applied the **SciPy** library to perform a **Delaunay Triangulation** to create a surface mesh of the lakebed
- Employed Blender's **Python** API to transform **2K+** points into a triangulated 3D model of the lake which allowed for exporting the model into a 3D print for an enhanced visualization of the lake
- Developed a website using **Three.js**, **Vite**, and **Bootstrap** to visually depict and interact with the 3D lake model, allowing users to view the surface and wireframe
- Presented findings as a presentation in the **Math Seminar** course and showcased a research poster at Stockton University's **NAMS Undergraduate Research Symposium**

Personal Projects

Finite Automata Designer

February 2025 – Present

- Built and deployed a full-stack web application using **Next.js** and **TypeScript** that lets users visually design, edit, and simulate **deterministic (DFA)** and **nondeterministic (NFA)** finite automata, hosted on **Vercel**
- Engineered an interactive **HTML5 Canvas** editor supporting drag-and-drop state creation, transition drawing, real-time rendering, custom hit-detection and a self-managed scene graph to track and manipulate automata
- Implemented Theory of Computation logic to simulate state transitions and validate input strings against user-built automata, returning acceptance and rejection results
- Integrated user **authentication** so users can save, load, and revisit their designs, and maintained an active open-source codebase of **700+** commits using **Git** issues and pull requests

Prayer Times

October 2025

- Developed a **Garmin** smartwatch application in Monkey C (Connect IQ) that calculates the five daily Islamic prayer times, sending **GPS** coordinates to a custom backend over HTTP and displaying the returned schedule
- Deployed the supporting **REST API** with **Node.js** and **Express** as a serverless function on **Vercel**, computing prayer times from latitude, longitude, and date with the open-source **PrayTimes.js** library
- Implemented protective controls for the public endpoint with **rate limiting** and **request throttling** (**express-rate-limit**, **express-slow-down**) to guard against abuse and keep responses reliable for the smartwatch client

Projects

DCGAN Animal Image Generation

April 2024 – May 2024

- Developed several basic Deep Convolutional Generative Adversarial Networks (**DCGANs**) in **PyTorch**, using **Batch Norm** and **ReLU** activations in generator and discriminator and optimizing training using **BCE** loss
- Preprocessed images from the Animal-10N and Kaggle Cats & Dogs datasets for image generation by performing resizing and normalization to ensure consistency and model compatibility
- Trained the best-performing model for **1,000** epochs on **12.5K** cat images, iterating on architecture and hyperparameters to improve output quality over earlier versions

Java Dynamic Website

May 2023 – June 2023

- Developed a dynamic **Java** web application applying **Apache Tomcat** as the application server to locally host the website and handle server-side requests.
- Enabled users to input simple polynomial and trigonometric expressions, evaluated and graphed using **exp4j** and **JFreeChart** APIs, with customizable x-axis bounds and graph resolution.
- Utilized **AJAX** requests to update the current page with a graph retrieved from the MySQL database without requiring a new page generation or triggering a refresh.